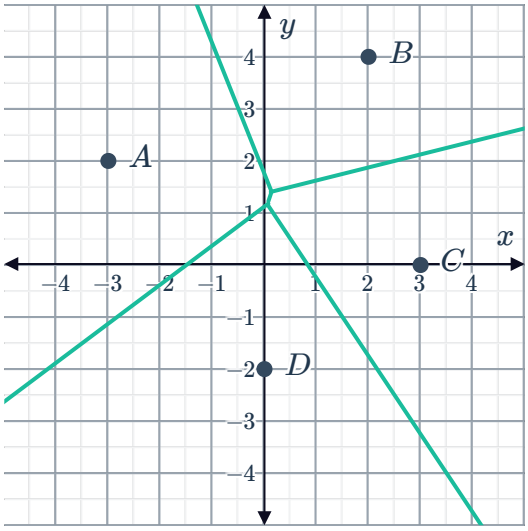
Nearest neighbour interpolation

1 Rainfall was measured at the following sites shown in the Voronoi diagram:



Site	Rainfall (mm)
A	10
B	6
C	8
D	17

Estimate the amount of rainfall at the following points:

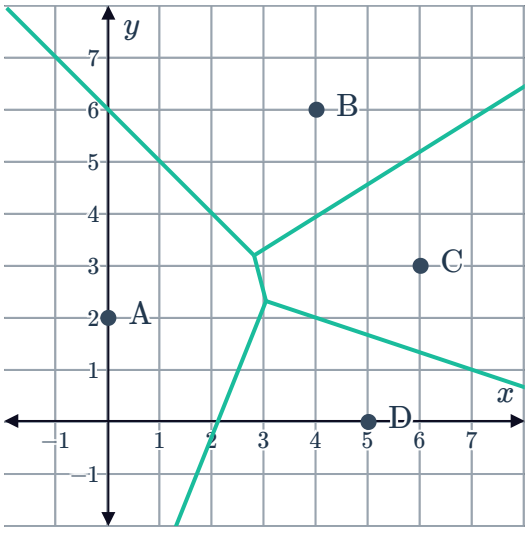
a

i $(2, 1)$ **ii** $(-3, -3)$

iii $(2.5, 2)$ **iv** $(-1.5, 0)$

b Would you expect it to rain more at $(3.5, -4)$ or $(0, 0)$? Explain your answer.

2 The following Voronoi diagram shows the locations of four weather stations in a city. The temperature at each station is shown in the table below:



Station	Temperature ˚C
A	24
B	18
C	20
D	16

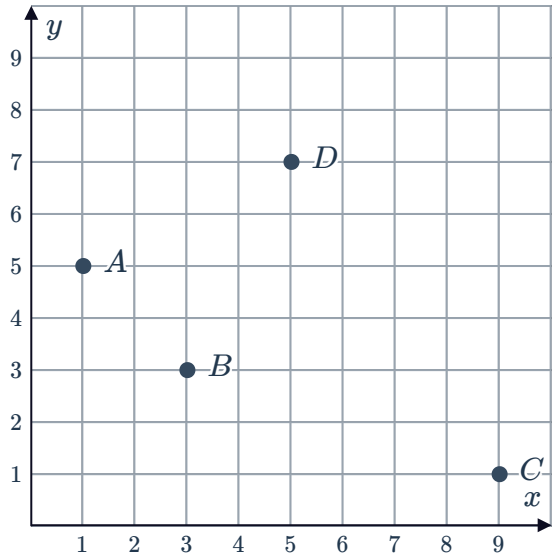
a Roland is having a party at his house which is located at $(1, 7)$ and wants to check the temperature. Estimate the temperature at Roland's house.

b Xinying lives at $(4, 2)$. Estimate the temperature at his house.

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3 The salinity of soil at a farm is measured at the locations shown on the following diagram. The level of salinity at each site is shown in the table below.

Site	Salinity level
A	4
B	6
C	8
D	12



a Construct a Voronoi diagram for the sites on the farm.

b The farmer wants to plant tomatoes in soil with salinity levels between 4 and 6 inclusive. State whether the farmer can plant tomatoes at the following locations:

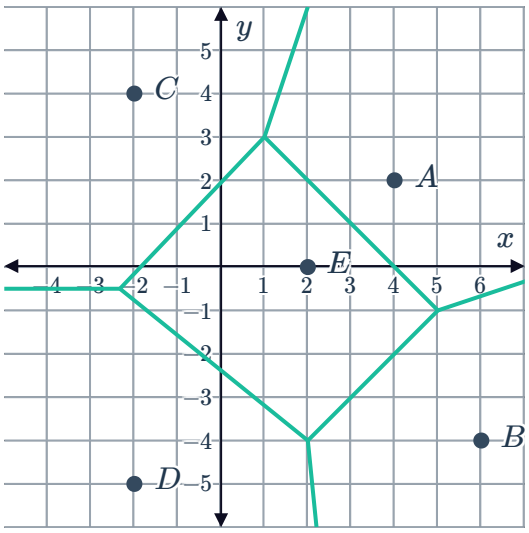
i $(2, 6)$ **ii** $(2, 8)$ **iii** $(5, 2)$ **iv** $(6, 2)$

c The farmer wants to plant spinach in soil with salinity levels between 6 and 10 inclusive. State whether the farmer can plant spinach at the following locations:

i $(1, 6)$ **ii** $(5, 4)$ **iii** $(7, 4)$ **iv** $(8, 1)$

4 The average depth of snowfall was measured at the five locations shown in the following Voronoi diagram during the month of December, where the units are in kilometres. The table below shows the results.

Site	Average snowfall (mm)
A	5
B	10
C	12
D	16
E	2



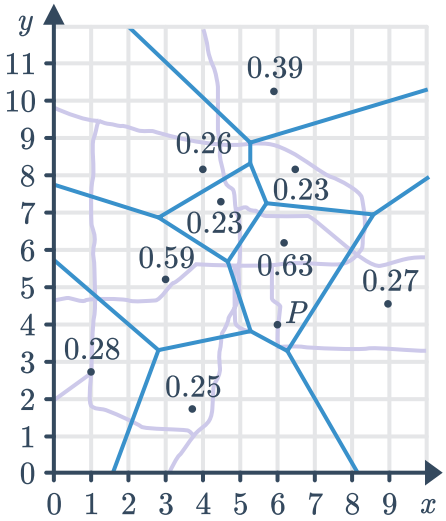
Hayley currently lives at $(-3, -4)$ and wants to move somewhere that does not snow too much. However she does not want to move more than 6 km away, since her son goes to public school near where she currently lives. Suggest the coordinates of a location that Hayley and her son could move to.

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5 The following Voronoi diagram shows the amount of rainfall measured on July 27, 2017 in Pueblo, Colorado, USA. The measurements show the average rainfall in each section, and are given in inches. Estimate the amount of rainfall at the following points:

a $(5, 3)$ **b** $(8, 4)$

c $(3, 10)$ **d** $(7, 7)$



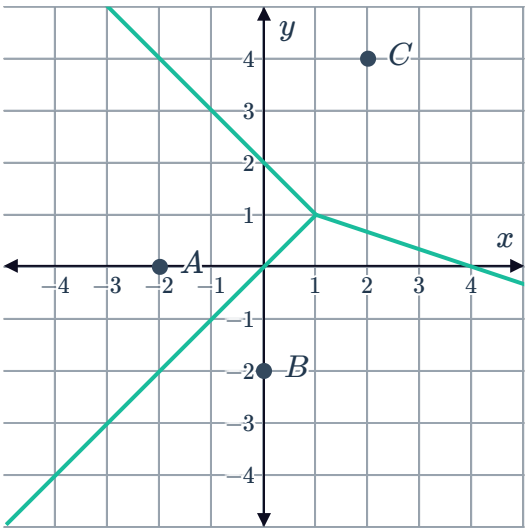
Largest empty circle

6 For each of the following Voronoi diagrams:

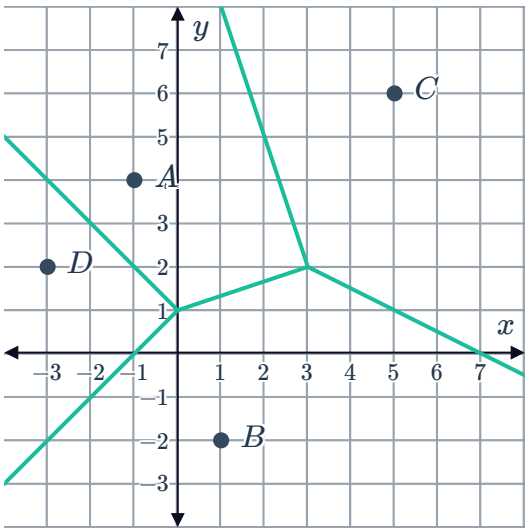
i Find the centre and radius of the largest empty circle. Round your answers to two decimal places.

ii Sketch the largest empty circle on the diagram.

a

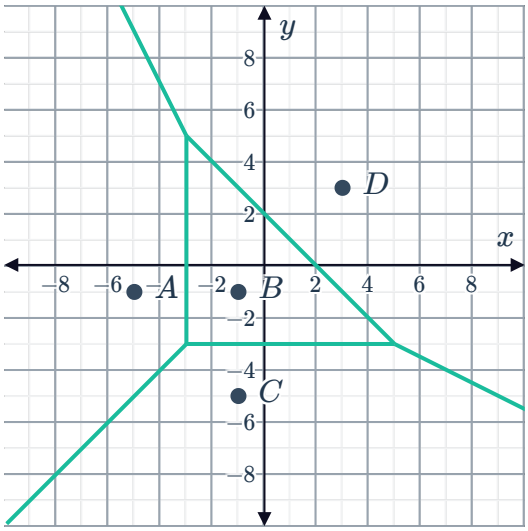


b



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7 Find the centre and exact radius of the largest empty circle on the following Voronoi diagram.



8 For each of the following sets of sites:

i Construct a Voronoi diagram.

ii Find the centre and radius of the largest empty circle on the diagram. Round your answers to three decimal places.

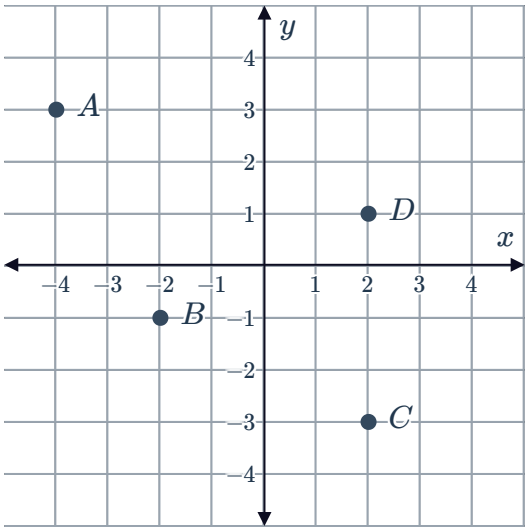
a $A(2,1), B(-4,3), C(2,-3)$

b $A(5,3), B(-1,5), C(-6,-1), D(3,-3)$

9 The following map shows the locations of four properties in a rural area:

a Construct a Voronoi diagram for these properties.

b A local site is going to become the location of a garbage tip. Find the coordinates of the location on the map of where the garbage tip should be placed so that it is as far away from the properties as possible.



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